Digital Chip Design Program

RTL Design

Roaa ALnawshan
Rawan aloyayri
Renad aloyayri
Lana alsuhaibani
Lara alsuhaibani
Rand Almutlaq

Task #6

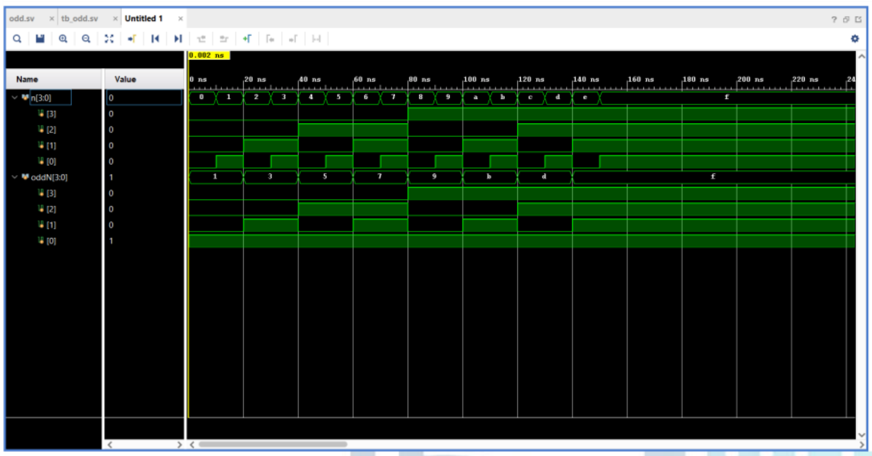
8/7/2026

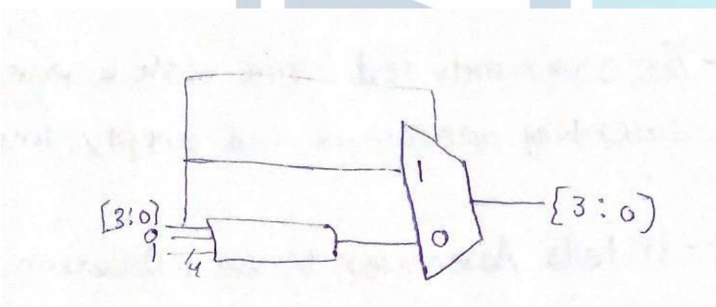
Combinational Datapath Design part 2

Task #1.1.1

```
module odd (
input logic [3:0] n,
output logic [3:0] oddN
);

assign oddN = (n[0] == 0) ? n + 1 :
n;

endmodule
```

```
`timescale 1ns/1ps

module tb_odd;

logic [3:0] n;
logic [3:0] oddN;

odd dut (.*);

initial begin
for (int i = 0; i < 16; i++) begin
n = i;
#10;
end

$finish;
end

endmodule
```

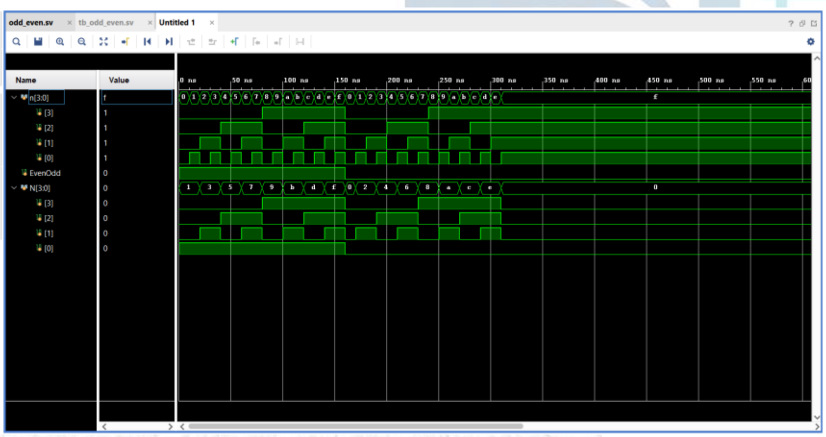


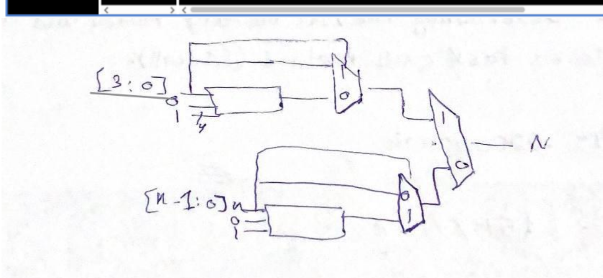


Task #1.1.2

```
module odd_even (
input logic [3:0] n,
input logic EvenOdd, // 1 = odd, 0 =
even
output logic [3:0] N
);

always_comb begin
if (EvenOdd) begin

if (n[0] == 0)
N = n + 1;
else
N = n;
end
else begin

if (n[0] == 1)
N = n + 1;
else
N = n;
end
end

endmodule
```

```
`timescale 1ns/1ps

module tb_odd_even;

logic [3:0] n;
logic EvenOdd;
logic [3:0] N;

odd_even dut (.*);

initial begin
EvenOdd = 1; // odd
for (int i = 0; i < 16; i++) begin
n = i;
#10;
end

EvenOdd = 0; // even
for (int i = 0; i < 16; i++) begin
n = i;
#10;
end

$finish;
end

endmodule
```

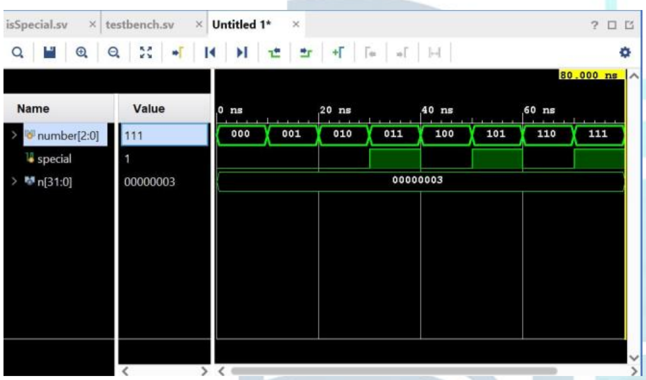


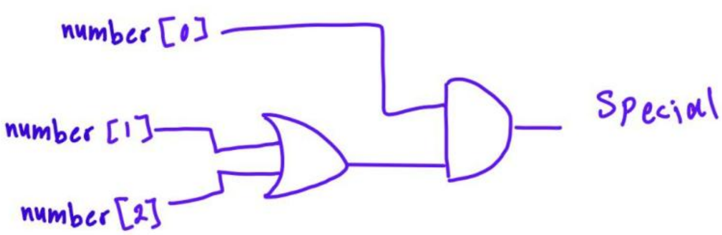


Task #1.2.1

```
module isSpecial
#(parameter n=3)
(
input logic [n-1:0] number ,
output logic special
);

assign special = number[0]&
(number[1]|number[2]);

endmodule
```

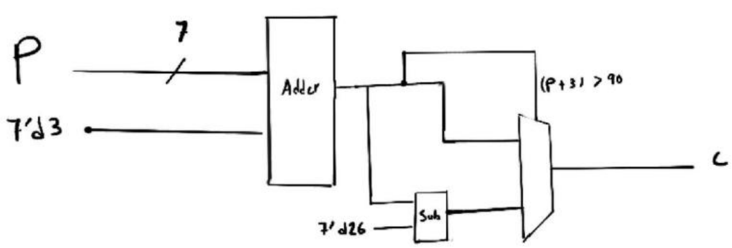
```
`timescale 1ns / 1ps
module testbench();
parameter n=3;

logic [n-1:0] number;
logic special ;

isSpecial #(.n(n))
dut(
.number(number),
.special(special)
);

initial begin
number = 3'b000; #10;
number = 3'b001; #10;
number = 3'b010; #10;
number = 3'b011; #10;
number = 3'b100; #10;
number = 3'b101; #10;
number = 3'b110; #10;
number = 3'b111; #10;

$finish;

end
endmodule
```





Task #1.3.1

```
module caesar_cipher (
input logic [6:0] p,
output logic [6:0] c
);

always_comb begin
if ((p + 7'd3) > 7'd90) begin
c = p + 7'd3 - 7'd26;
end
else begin
c = p + 7'd3;
end
end
```




```
`timescale 1ns/1ps

module testbench();

logic [6:0] p;
logic [6:0] c;

caesar_cipher dut (
.p(p),
.c(c)
);

initial begin
p = 7'd65;
#10;

p = 7'd90;
#10;

$finish;
end

endmodule
```

Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report	✓					
Task 1.1	✓					
Task 1.2					✓	✓
Task 1.3			✓	✓		